

James Fox

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Senior AI Applied Innovation Specialist with 7 years of full-stack software development experience in pharmaceutical R&D. Proven track record of working closely with scientific stakeholders for consultation, requirements gathering and feedback. Experience managing servers and using cloud-based environments. Keen to apply my skills to new problems, continually learn and broaden my experience.

Core Technical Skills

- **Languages & Frameworks:** JS/TS (Vanilla, React/NextJS), Java, Python.
- **AI & Data Science:** LLM Integration (RAG, Claude plugins / skills), python ML and data science libraries (Pandas, SciKit-learn, numpy).
- **Infrastructure & Architecture:** Docker, gRPC, REST APIs, cloud-based systems (AWS), GxP server maintenance, three-tier architecture.
- **Databases:** Relational (Oracle & PostgreSQL), self-taught NoSQL (Neo4j) experience.

Employment

AstraZeneca, Cambridge (September 2019 – Present)

Senior AI Applied Innovation Specialist | Enterprise AI (June 2026 – Present)

- Move as a result of reorg to Enterprise AI unit, focusing on building suites of LLM tools to support multiple business areas in AZ ambition to optimise processes 30% by 2030.
- Built a RAG project from scratch as a learning opportunity. Solving a key problem of visibility across AZ. Resulted in a tool which allows employees to find projects and subject matter experts in the company with a simple plain text query. Generated ongoing collaborative effort with Senior Director and others.
- Currently developing a ‘work intelligence’ platform. Ultimately a decision graph aligned against individual/project/company goals enabling searchable reasoning for project direction and outcomes, and guided decision-making for the future.
- Took part in a spike project to develop plugins for Claude Cowork, intended for use by C-suite / SET. Particular interest in ensuring stability and security of these tools.

Senior Data Scientist, Informatics | BioPharma R&D (November 2023 – June 2026)

- In-role promotion. Continued focus on full-stack dev for early R&D teams.
- Extended development of RNAopt, a bespoke, in-house tool to optimise mRNA for expression in various therapeutic modalities. Added batch processing and REST API for decoupled and API-driven submissions.
- Core achievement: Bootstrapped RNAopt with literature-best tool to create a 3D Pareto frontier of optimal sequences, in a non-constant-sum multi-objective trade-off space. Developed algorithm to maintain RNAopt-optimised secondary structure and hone sequence using literature-best outputs, proved superior expression. **Runner-up in AZ R&D awards. Patent pending.**
- Direct line-management of L7 Bioinformatics Apprentice and University Industrial Placement Student.
- Successfully tripled size of team, leading recruitment following ‘relocation strategy’ focusing on expanding footprint in Barcelona. Recognised for ‘exceptional contribution’ in reward window.
- Responsible for app server maintenance, cert renewals and PoC for other developers encountering deployment issues. Took technical ownership of GxP server supporting clinical work and infectious disease vigilance.

Data Scientist, Informatics | BioPharma R&D (March 2022 – November 2023)

- Hired full-time into team post-apprenticeship.
- Focused work on web-app development using gRPC framework and servlets.

- Deployed and maintain a thread-safe web-app allowing scientists to run computationally expensive sequence optimisation on demand.
- Selected by the head of department to be a social lead for the group. Successfully connected high-profile stakeholders together for end of year celebration of our department's work.
- Successfully created and sustained the AZ Apprentice Social Society, acting as chair with a mission to create a diverse and unified family of apprentices.

***L7 Apprentice Bioinformatics Scientist* | BioPharma R&D (September 2019 – March 2022)**

- Work within a support group to deliver applications to lab scientists, both simplifying and automating workflow processes.
- Successful deployment of first independent app a year into role, a large project to graphically derive lineage of therapeutics from captured data.
- Treasurer of apprentice charity committee, Science Industry Partnership taskforce member overseeing the delivery of careers outreach activity to young talent and lobbying for greater investment.

GlaxoSmithKline, Worthing (July 2017- July 2018)

***Fermentation Scientist* (Industrial Placement)**

- Developed a calculator to determine feed rates of downscaled processes based on fermentation conditions, significantly reducing time spent, and error caused by, manually fine-tuning feed rates.
- Voluntary work at careers fairs and Big Bang Science Fair, Birmingham, as company representative.

Education

Applied Bioinformatics MSc (p/t): Cranfield University (Sept 2019 – Sept 2021)

Pass with distinction – 85% overall including 94% and 88% in Python and Java modules respectively. Project-lead of a 10-week group project providing vital improvements to *MapOptics*, a genome assembly pipeline tool that achieved 87% overall.

Biological Sciences BSc: Bournemouth University (Sept 2015- June 2019)

First-Class - 84% overall

A Levels/ AS: Hills Road Sixth Form College, Cambridge (Sept 2013- June 2015)

Biology, Mathematics, Chemistry

GCSEs: Bassingbourn Village College (Sept 2008- June 2013)

10 GCSEs including Maths, Science and English

Interests

- **Home Infrastructure & Tech:** Recently set up a home server running Ubuntu Server and a Docker stack to host multiple applications. Maintain an active interest in machine learning, AI, and emerging consumer products. Started learning programming at university from online courses and content.
- **Literature:** Enjoy fantasy, scientific, historical and philosophical non-fiction.
- **Sports & Outdoors:** Member of local squash club, active runner and cyclist. Annually camp and hike in locations such as Lake District and Snowdonia. Recently returned from a trip to Dolomites walking section of Via Alta 1.